Workshop Setup 🚀

Completing this setup before the workshop ensures we can dive straight into coding without install delays.

Prerequisites

- **Git:** https://git-scm.com/downloads
- **uv (package manager):** https://docs.astral.sh/uv/getting-started/installation/
  Used to manage project dependencies and ensure consistent Python versions. After installing `uv`, restart the terminal.
- **A code editor/IDE:** For example, VS Code: https://code.visualstudio.com/download
- **A way to work with Jupyter Notebooks:** For example, the Jupyter extension for VS Code: https://code.visualstudio.com/docs/datascience/jupyter-notebooks

Step 1: Clone the course repo

Run:

```
git clone https://github.com/arilamstein/streamlit-workshop.git
cd streamlit-workshop
```

github.com/arilamstein/streamlit-workshop

Please do the setup if you have not done so already.

Building Data Apps In Python With Streamlit

Session 1: Intro

Ari Lamstein
AriLamstein.com

Exercise: What is a Data App?

Type your answer to these 2 questions in the chat:

1. What do you already know about Data Apps and Streamlit?
2. What do you want to build after completing this course?

About me

Ari Lamstein

- Former software engineer in San Francisco (~10 yrs)
- Open-source contributor & Trainer (R)
- Transitioned to Python in recent years
- Built civic data apps with Streamlit: *Covid Demographics Explorer* and *Immigration Enforcement Explorer*

Before I give my own answer about what a "data app" is, let me introduce myself

Exercise: What is a Data App?

Type your answer to these 2 questions in the chat:

1. What do you already know about Data Apps and Streamlit?
   - Data Apps vs. Jupyter Notebooks
   - Streamlit Apps require very little code
2. What do you want to build after completing this course?

I have two answers to the "What is a data app?" question.

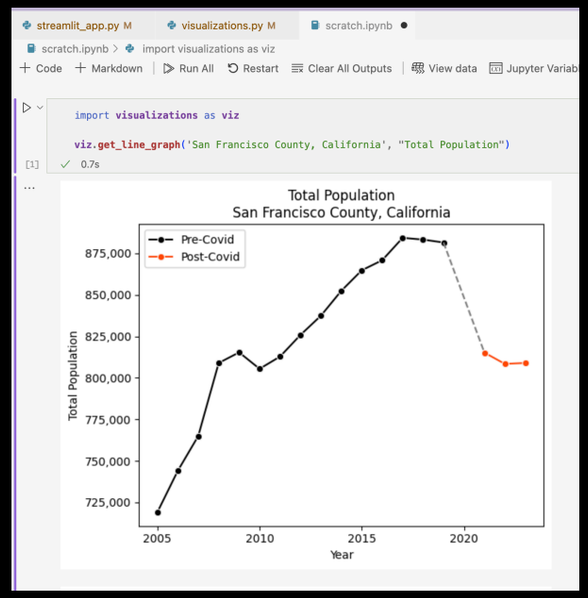
Data Apps vs. Jupyter Notebooks

- Jupyter Notebooks let me create a single story about a dataset. I can share the notebook with others for free via github.
- Streamlit lets me create an app to explore a dataset. I can share the app with others for free via Streamlit Cloud.
- I often start with a notebook, and end with a Streamlit app.

Ask: How many people here have used Jupyter notebooks before?
In my own work, I view Streamlit as a companion to Jupyter notebooks.

Example Jupyter Notebook

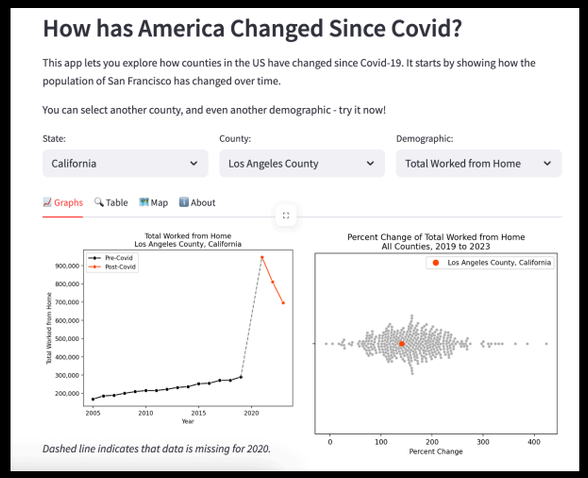
- This notebook tells a story about one city.



At one point I was interested in understanding how San Francisco's population changed during Covid. My first step was to create a Jupyter Notebook that had graphs like this.

Example Streamlit App

- This Streamlit App lets users see how the population of their county changed
- You'll be creating a similar app today



Note the interactivity:

-select boxes -> graphs change.

-layout: tabs are clickable

The app you'll be creating today is similar to this. But it will focus on states instead of counties.

Streamlit Apps: very little code

- Open the file streamlit_app.py in the directory 1-intro
- Just 4 lines of code!
  - import streamlit as st
  - st.dataframe

session1 > streamlit_app.py > ...

```python
import streamlit as st
import pandas as pd

#st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

st.dataframe(df)
```

Please open the code for the demo app now. This is what you should see: it's just 4 lines of code! And two of those lines are imports. The only real "streamlit" part of it is the st.dataframe call - that's how we get the dataframe to appear in the app. Streamlit apps are just normal Python programs that import the streamlit library.

Learning Goals: Intro

1. Exercise: "Rerun" a Streamlit App
2. Exercise: Running a Jupyter Notebook

The goal of this part of the course is to get all the prerequities out of the way so you can dig into the core parts of Streamlit. To that end, we're going to do 2 exercises: the first is to learn how to "rerun" a streamlit app.

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```

github.com/arilamstein/streamlit-workshop

I started the workshop by asking everyone if they had done the setup. From this point on, I'm going to assume that everyone has at least cloned the repo, installed uv and created the virtual environment. Does anyone need help with that?

Activate the Virtual Environment

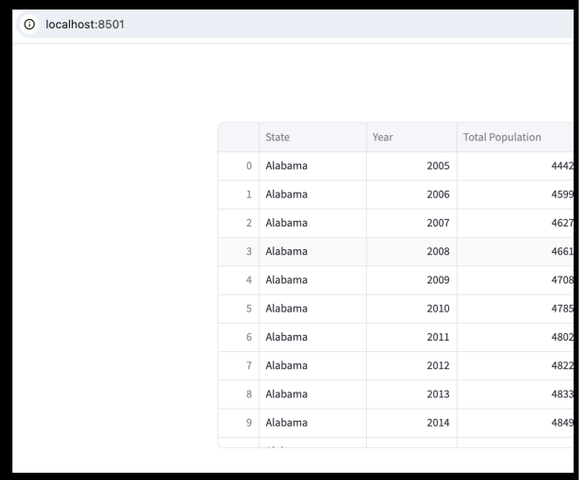
- Activate the virtual environment:
  - **Mac/Linux**: source .venv/bin/activate
  - **Windows (Command Prompt)**: .venv\Scripts\activate
  - **Windows (PowerShell):** .venv\Scripts\Activate.ps1

- Test that the virtual environment is activated:
  - Run python --version → should report a version starting with **Python 3.13**
  - Run which python (Mac/Linux) or where python (Windows) → should show a path inside **.venv**

The next step is to activate your virtual environment. The command to activate the virtual environment depends on your operating system. Type the right command now. And then type "python --version" to make sure that you have the right version of Python installed.

Run the Demo App

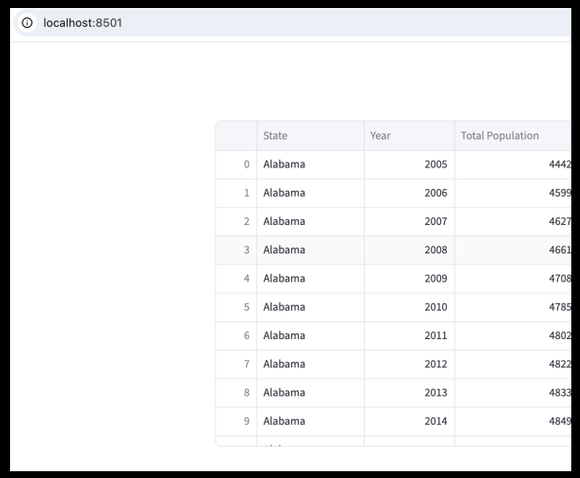
- Each session has its own directory
- Type `cd 1-intro`
  - This moves to the directory for this session
- Type `streamlit run streamlit_app.py`
  - This runs the demo app for this section



Move to the "1-intro" directory and type "streamlit run streamlit_app.py".
It should open a browser like this.

Running the App

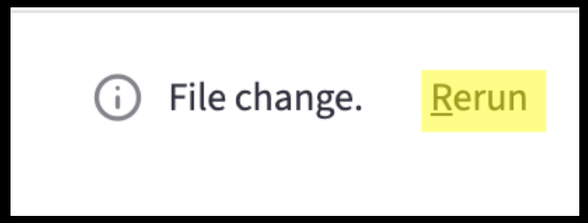
- It should open a browser like this
- Click the table headers to sort (e.g. State, Year, Total Population)
- **Quiz:**
  - What are the min and max years of the dataset?
  - What state had the lowest population?
  - What state had the highest income?



	State	Year	Total Population
0	Alabama	2005	4442
1	Alabama	2006	4599
2	Alabama	2007	4627
3	Alabama	2008	4661
4	Alabama	2009	4708
5	Alabama	2010	4785
6	Alabama	2011	4802
7	Alabama	2012	4822
8	Alabama	2013	4833
9	Alabama	2014	4849

Rerun

- Streamlit is designed for fast iteration
- Each time you modify and save the script the word Rerun will appear in the top right of the app
- Click it to rerun the app with your changes
- Do not stop and restart the app from the command line



In the next exercise I'm going to ask you to make a small change to the app and then rerun it.

Exercise: Modify & Rerun

- To start:
  - Open the file streamlit_app.py in the directory 1-intro
  - Type streamlit run streamlit_app.py to run the app
- Then:
  - Uncomment the st.title line
  - Save the file
  - Click "rerun" and verify that the title appears

session1 > streamlit_app.py > ...

```python
import streamlit as st
import pandas as pd

#st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

st.dataframe(df)
```

This is an exercise to make sure you're comfortable changing the app, and clicking "rerun" to see your changes.

Result

Demo Streamlit App

	State	Year	Total Population	Median Household Income
0	Alabama	2005	4442558	36879
1	Alabama	2006	4599030	38783
2	Alabama	2007	4627851	40554
3	Alabama	2008	4661900	42666
4	Alabama	2009	4708708	40489
5	Alabama	2010	4785298	40474
6	Alabama	2011	4802740	41415
7	Alabama	2012	4822023	41574
8	Alabama	2013	4833722	42849
9	Alabama	2014	4849377	42830

You should see this screen. Now there's a new title.

Streamlit

- Lets the script run as a webapp
- Imported as st
- Two output functions:
  - st.title
  - st.dataframe
- Processed top to bottom.
- Cannot run in a Jupyter notebook - must be a regular .py file.

session1 > streamlit_app.py > ...

```python
import streamlit as st
import pandas as pd

st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

st.dataframe(df)
```

Demo Streamlit App

	State	Year	Total Population	Median Household Income
0	Alabama	2005	4442558	36879
1	Alabama	2006	4599030	38783
2	Alabama	2007	4627851	40554

Review: What is Streamlit? It's a library that let's you turn Python Scripts into web apps. Each time you want something to appear in the web app, you call a function in the streamlit library (st.title, st.dataframe), and pass in the object you want to appear. The order in which things appear in the file is the order in which they appear in the app (ex. title appears above table)

Learning Goals: Intro

1. Exercise: "Rerun" a Streamlit App
2. Exercise: Running a Jupyter Notebook

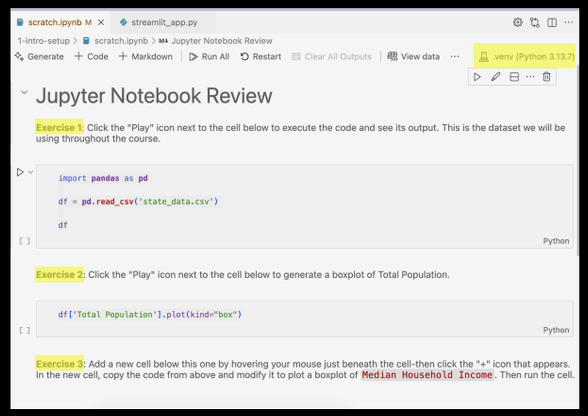
We're going to be using Jupyter Notebooks throughout this course. So let's do a short exercise to make sure that you can run the notebooks in the repo.

Exercise: Jupyter Notebooks

- Open review.ipynb. Two ways:
  1. VS Code with Jupyter plugin
  2. Jupyter notebook via command line:

     uv run jupyter notebook scratch.ipynb

- Complete the 3 exercises as a review



Later on we'll be using Jupyter Notebooks. I want to include with notebooks now to make sure everyone is comfortable with them. Note that in VS Code you'll need to select a "kernel" in the upper right hand corner.

Section 1 Review

- What is a data app?
- Confirm setup:
  - Clone course repo
  - Create virtual environment with uv
  - Activate virtual environment
- 2 Exercises:
  - Modify & Rerun app
  - Jupyter Notebooks

The goal of this module was to get you to a point where you can dig into the core of Streamlit.